

What to Know Before Starting

Use this document to accompany the second introduction video for the "AI Agents for Everyday Professionals" course.

Before diving into building AI agent workflows, it's essential to understand the key concepts and terminology. This guide will familiarize you with the fundamental terms you'll encounter throughout the course, ensuring you can confidently navigate AI agent platforms and understand how these automated workflows function.

Core Concepts: Understanding AI Agents

AI Agent vs. Large Language Model

An **AI agent** is fundamentally different from a large language model (LLM) like Claude, ChatGPT, or Gemini. While an LLM is a generalist that can be passively prompted to follow instructions, an AI agent workflow is actively created to specifically execute instructions on your behalf in a repeatable, scalable, and consistently reliable manner.

Example: It may be difficult to seamlessly prompt a large language model to log into a specific account, book a flight, send you the itinerary, and summarize all that information in a neatly packaged PDF. However, with an AI agent workflow, you can set this up as an automated, repeatable process that executes reliably every time.

Think of AI agents as little taskmasters that have mastered a very specific workflow from start to finish and need to be initiated with a specific trigger event.

AI Agent Workflows vs. AI Agent Mode

There's an important distinction between building AI agent workflows (what we do in this course) and the AI agent mode you might see in tools like ChatGPT. The key difference is **precision**.

AI Agent Mode in ChatGPT: This feature automates the entire process by having an agent come up with what it thinks is an accurate workflow for the tasks you're trying to achieve, then executes that workflow on your behalf. However, there's significant

room for error, hallucination, and mistakes in this automated approach.

Custom AI Agent Workflows (using N8N): By precisely creating AI agent workflows step-by-step, we can mitigate errors and ensure reliable execution. You maintain full control over each step, reducing the risk of hallucinations and unexpected behavior.

Essential Terminology

Workflow

A **workflow** is a series of steps that are chained together in order to complete a task. In productivity tools, workflows are typically illustrated as a series of connected nodes that represent different actions or processes that must occur in sequence.

You'll see workflows visualized in the N8N interface as connected boxes (nodes) that flow from one step to the next, creating a clear map of how your automated task progresses from start to finish.

Node

A **node** is a building block in your workflow. Each node usually contains a series of instructions that must be completed in order for the task to move forward to the next step. Nodes always appear as little boxes in the workflow interface.

Each node performs a specific function within your workflow—it might extract data from a file, send an email, call an AI model, or perform a calculation. The output of one node typically becomes the input for the next node in the sequence.

Trigger

A **trigger** is a specific type of node that has certain conditions that must be met in order to start the execution of the workflow. Triggers are always the first step in any AI agent workflow—they define what event will kick off your automation.

Common trigger examples:

- Uploading a spreadsheet to a specific location
- Receiving an email in a specific inbox
- Submitting a form with specific data
- A scheduled time or date being reached

- A webhook being called by another application

API (Application Programming Interface)

An **API** is a way for different software applications to communicate with each other. In the context of AI agent workflows, you'll most commonly use large language model APIs like the APIs for ChatGPT, Claude, or Gemini.

When you integrate an LLM API into your workflow, you're essentially using the brain of that AI model to do the thinking on behalf of your AI agent workflow. The API allows your workflow to send information to the AI model and receive intelligent responses back.

API Key

An **API key** is a unique code that you generate to access the capabilities of a large language model within your AI agent tools. It acts as both an authentication method and a way to track your usage of the API.

You can generate an API key through the developer platform of whichever large language model you choose to use (ChatGPT, Claude, Gemini, etc.). This key is then entered into N8N to give your workflow access to that AI model's capabilities.

Putting It All Together

Understanding how these concepts work together is key to building effective AI agent workflows. Here's how the pieces fit:

1. **You design a workflow** that represents the complete process you want to automate
2. **The workflow consists of multiple nodes**, each performing a specific task in sequence
3. **A trigger node initiates the workflow** when specific conditions are met
4. **One or more nodes use AI APIs** (via API keys) to add intelligent processing to your workflow
5. **The workflow executes automatically**, completing your task reliably and consistently

Complete Example: You create a workflow where the trigger is uploading a PDF contract. The first node extracts text from the PDF. The second node uses Claude's API (authenticated

with your API key) to analyze the contract and identify negotiation points. The final node sends you an email with the analysis. Each time you upload a contract, this entire workflow executes automatically.

What Makes This Approach Powerful

- ✓ **Repeatability:** Once built, the workflow runs the same way every time
- ✓ **Scalability:** Handle one task or thousands with the same workflow
- ✓ **Reliability:** Precise control reduces errors and hallucinations
- ✓ **Consistency:** Every execution follows your exact specifications
- ✓ **Transparency:** You can see and understand each step in the process

Key Takeaways

Before building your first AI agent workflow, remember these foundational concepts:

- **AI agents are specialists**, not generalists—they excel at specific, well-defined workflows
- **Workflows are chains of nodes** that execute in sequence to complete a task
- **Triggers initiate workflows** based on specific conditions being met
- **APIs connect your workflow to AI capabilities**, requiring API keys for access
- **Precision-built workflows reduce errors** compared to AI agent modes that automate workflow creation

With this foundational understanding, you're ready to start building your own AI agent workflows. The remaining terms and concepts will be introduced as you progress through hands-on examples in the upcoming videos.